

Data Processing & Business Analysis Orientation

Session Objective

Understand:

- Basics of IT
 - How IT works in businesses
 - ERP & CRM systems
 - IT job roles and skills
 - Data Processing & Business Analysis course
 - Career opportunities after course completion
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1. Introduction to Information Technology

What is IT?

Information Technology (IT) is the use of computers, software, databases, networks, and the internet to collect, store, process, manage, and share information in a fast and efficient way.

IT helps individuals and organizations perform their daily activities more easily, accurately, and quickly by using technology instead of doing everything manually.

Examples

- Online shopping
- ATM transactions
- Hospital management systems
- Mobile banking

Components of IT

- Hardware → Computer, laptop
- Software → applications
- Database → Stores data
- Network → Internet

Real-Time Example

In hospitals:

- Patient data stored digitally
 - Online appointments
 - Billing generated automatically
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2. Role of IT in Business Operations (20 Minutes)

How IT Helps Businesses

- Saves time
- Reduces manual work
- Improves accuracy
- Generates reports quickly
- Helps management make decisions

Departments Using IT

- HR
- Finance
- Sales
- Inventory
- Customer support

Example

When a customer places an order:

- Stock updates automatically
 - Bill is generated
 - Delivery tracking starts
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1. ERP (Enterprise Resource Planning)

ERP is used to manage the internal operations of a company in one system.

ERP Handles

- Employee data
- Finance
- Inventory
- Sales
- Purchase

Real-Time Example – Hospital

A patient comes to hospital:

1. Reception creates patient entry
2. Doctor updates treatment details
3. Pharmacy updates medicine stock
4. Finance generates bill

All departments work in one ERP system.

Result

- Faster work
- Less paperwork
- Accurate reports

Popular ERP Tools

- SAP ERP
- Oracle ERP

2. CRM (Customer Relationship Management)

CRM is used to manage customer interactions and improve customer service.

CRM Stores

- Customer details
- Calls and emails
- Complaints
- Purchase history

Real-Time Example – E-commerce Company

A customer orders a mobile phone.

Using CRM:

- Company stores customer details
- Tracks customer complaints
- Sends offers and updates
- Customer support team checks order history

Result

- Better customer relationship
- Improved customer satisfaction
- Increased sales

Popular CRM Tools

- Salesforce
- Zoho CRM

-APIs (Application Programming Interfaces)

API is a connection that allows two applications to communicate with each other.

Simple Example

When you use a food delivery app:

- App sends request to restaurant system
- Restaurant sends menu and order status back

This communication happens using APIs.

Real-Time Examples of APIs

Payment API

Used in:

- Google Pay
- PhonePe
- Online shopping websites

Helps process payments securely.

Weather API

Weather apps get live weather data from another server using APIs.

Google Maps API

Apps use it to:

- Show locations
 - Track deliveries
 - Find routes
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Benefits of APIs

- Faster data sharing
 - System integration
 - Automation
 - Real-time communication between applications
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4. Interplay Between IT and Business Strategy (15 Minutes)

Business Strategy

Company plans to grow and improve profits.

How IT Supports Strategy

- Faster work
- Better customer service
- Data-based decisions
- Automation

Example

Online shopping companies use:

- Websites
- Databases
- Delivery tracking systems
- Analytics dashboards

Result:

- Better customer satisfaction
 - Increased sales
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5. IT Industry Job Roles & Skills Required (25 Minutes)

Job Role	Work	Skills Required
Data Analyst	Analyze data & reports	Excel, SQL, Power BI
Business Analyst	Understand business needs	Communication, Excel
SQL Developer	Work with databases	SQL, DB concepts
Software Developer	Build applications	Java,python, API's etc....
System Administrator	Manage systems	Networking, Troubleshooting

6. About Our Course – Data Processing & Business Analysis (15 Minutes)

What You Will Learn

MS Excel

- Formulas
- Pivot tables
- Charts
- Data cleaning

SQL

- Databases
- Queries
- Data filtering

Power BI

- Dashboards

- Visualization
- Reports

Business Analysis Basics

- Requirement gathering
- Understanding business problems

Basic Statistics

- Data analysis
 - Trends and patterns
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7. Course Outcomes & Career Opportunities (10 Minutes)

Outcomes After Course

Students can:

- Analyze business data
- Create dashboards
- Generate reports
- Work with databases
- Support business decisions

Job Roles After Completion

- Data Analyst
- MIS Executive
- Reporting Analyst
- Junior Business Analyst
- Power BI Developer
- SQL Analyst

Industries Hiring

- Healthcare

- Banking
 - Retail
 - IT companies
 - E-commerce
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Final Conclusion (5 Minutes)

Key Takeaways

- IT is used in every business today
- Data is very important for companies
- Excel, SQL, and Power BI are highly useful skills
- Business Analysis helps solve business problems
- This course opens opportunities in IT and analytics careers

Final Message

Learning data processing and business analysis helps students build strong careers in the growing IT industry.